

The Computational Complexity of Binary Phase-Mask Design

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We study the combinatorial optimization problem that appears in the design of binary phase masks. We prove by reduction from the CLIQUE decision problem on graphs that despite its apparent simplicity and structure, the optimization problem is NP-hard.

I. INTRODUCTION

The design of phase masks that match a prescribed intensity output pattern is an important inverse problem in optics with many applications across science and engineering. Typically, these problems are solved with iterative algorithms. The Gerchberg-Saxton algorithm [1] is a historically important early example of this. Iterative methods of this kind generally do not provide guarantees for optimality, runtime or precision for the type of discrete design problem considered here. The performance of such algorithms can also depend sensitively on initialization.

In this paper, we study the nature of these optimization problems by focusing on the design of binary phase masks. We demonstrate that this version of the problem is NP-hard.

Previous works have established the computational hardness of related phase-retrieval problems. Fickus et al. [2] have shown that phase retrieval from certain almost injective measurements is NP-hard. In recent work, Huang and Xu [3] have proved a hardness result for one-dimensional Fourier phase retrieval.

In contrast to these works, we focus here on the inverse-design problem with fixed Fourier operator and the signal constrained to two possible phase settings.

Our proof begins by rewriting the cost function of the optimization in terms of four parts, one of which involves circulant matrices. We next describe how to encode into this part any circulant adjacency matrix by choosing the target intensity patterns of the phase mask. This allows us to establish a connection to computational problems in graph theory. In particular, we show that our optimization problem encodes as a special case the CLIQUE problem: decide if there is a clique of size K in the graph. This famous decision problem is known to be NP-complete on general graphs. In our proof we make heavy use of the results obtained by Codenotti, Gerace and Vigna [4], who used Bose-Chowla sets to prove that CLIQUE remains NP-complete when restricted to circulant graphs, i.e. graphs with circulant adjacency matrix.

II. INTRODUCTION TO THE BINARY PHASE-MASK DESIGN PROBLEM

We consider the problem of designing a one-dimensional binary phase mask that consists of N pixels. Starting from the

discretized Fraunhofer diffraction model, we assume that the light beam falling onto pixel i will experience a phase-shift equal to either one of two complex phases $c_1, c_2 \in \mathbb{C}$ with $|c_1| = |c_2| = 1$. We introduce a binary variable vector $\mathbf{z}$ with components $z_i \in \{0, 1\}$ that selects the phase setting of each pixel. Then the output signal $\mathbf{x}$ leaving the diffraction grating is given by

$$\mathbf{x}(\mathbf{z}) = c_1 \mathbf{1} + (c_2 - c_1) \mathbf{z}, \quad (1)$$

where $\mathbf{1}$ is the vector with all entries equal to 1. Since multiplication of the phase-mask with a global phase factor does not affect the output intensity vector, we will set in the following $c_1 = 1$ and write $c_2 = \exp(i\gamma)$.

In the Fraunhofer limit, the output signal $\mathbf{y}$ is obtained by Fourier transforming $\mathbf{x}$, i.e.

$$\mathbf{y} = F \mathbf{x}, \quad (2)$$

where the Fourier matrix F is the $N \times N$ unitary matrix with entries

$$F_{jk} = \frac{\omega^{jk}}{\sqrt{N}}, \quad 0 \leq j, k < N \quad (3)$$

$$\omega = e^{2\pi i/N} \quad (4)$$

We consider the following design problem for the phase masks. Given an intensity vector $\mathbf{b}$, design a signal intensity vector $\bar{\mathbf{y}} \odot \mathbf{y}$ that is as close as possible to $\mathbf{b}$. Here, $\odot$ denotes the elementwise multiplication of vector components. Since the input intensity is $|\mathbf{x}|_2^2 = N$ and F is a unitary matrix, Parseval's identity yields $|\mathbf{y}|_2^2 = N$. Thus we consider target intensity vectors that satisfy $\sum_i b_i = N$ and $b_i \geq 0$. These are reasonable physical constraints. In fact, the reduction below constructs target intensity vectors satisfying these. In other words, the optimization remains NP-hard even with such constraints.

We measure closeness by the L^2 norm, thus the objective function is

$$C(\mathbf{z}) = \|\bar{\mathbf{y}} \odot \mathbf{y} - \mathbf{b}\|_2^2 \quad (5)$$

$$= \|\overline{F \mathbf{x}(\mathbf{z})} \odot F \mathbf{x}(\mathbf{z}) - \mathbf{b}\|_2^2 \quad (6)$$

and the optimization problem can be formulated as

$$\min_{\mathbf{z} \in \{0,1\}^N} C(\mathbf{z}). \quad (7)$$

The function $C(\mathbf{z})$ is a particular subclass of degree-4 pseudo-boolean functions. This is seen upon expanding the square in

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Eq. (6). The objective function becomes the sum of several simple parts:

$$C'(z) = C_4(z) + C_3(z) + C_2(z) + C_1(z) + C_0 \quad (8)$$

which are respectively the quartic, cubic, quadratic, linear and constant term. The constant C_0 will be ignored from here on since it does not affect the minimizer vector z .

The quartic term has the form

$$C_4(z) = \frac{16}{N} \sin^4\left(\frac{\gamma}{2}\right) \sum_{j,k,l} z_k z_{j+k} z_l z_{j+l}, \quad (9)$$

where all summations are understood, here and below, as ranging over all of $\mathbb{Z}/N\mathbb{Z}$ – the integers modulo N . Plainly, in this term the intensity vector $\mathbf{b}$ is not even present.

Similarly, in the cubic term

$$C_3(z) = -\frac{32}{N} \sin^4\left(\frac{\gamma}{2}\right) H^3 \quad (10)$$

the intensity vector does not enter. Here, H is the Hamming weight of z :

$$H = \sum_i z_i \quad (11)$$

The quadratic term consists of two parts, one of which contains the intensity vector explicitly:

$$C_2(z) = -8 \sin^2\left(\frac{\gamma}{2}\right) \sum_{j,k} M_{jk} z_j z_k \quad (12)$$

$$+ 8 \sin^2\left(\frac{\gamma}{2}\right) \left[1 + 2 \sin^2\left(\frac{\gamma}{2}\right)\right] H^2, \quad (13)$$

where the matrix M is given by

$$M_{jk} = \sum_{n=0}^{N-1} c_n \cos\left(\frac{2\pi}{N} n(j-k)\right) \quad (14)$$

$$c_n = \frac{1}{N} b_n \in \mathbb{R}_{\geq 0}. \quad (15)$$

The first term in $C_2(z)$ depends on the intensity vector and allows us to encode adjacency matrices of graphs. This becomes the source of the NP-hardness of the optimization problem.

Finally, the linear term is

$$C_1(z) = -8 \sin^2\left(\frac{\gamma}{2}\right) (N - b_0) H \quad (16)$$

where only b_0 , the first component of the intensity vector, appears explicitly.

Thus the full optimization is an unconstrained binary optimization problem

$$\min_{z \in \{0,1\}^N} C_4(z) + C_3(z) + C_2(z) + C_1(z). \quad (17)$$

III. BINARY PHASE-MASK DESIGN IS NP-HARD

We show that the optimization task in Eq. (17) is NP-hard. We prove this by establishing a connection with graphs.

Theorem 1. *For every fixed $\gamma \not\equiv 0 \pmod{2\pi}$, the optimization problem in Eq. (7), with $c_1 = 1$ and $c_2 = e^{i\gamma}$, is NP-hard.*

Equivalently, the unconstrained binary optimization problem in Eq. (17) is NP-hard. It turns out that an algorithm that solves our optimization problem can also be used to decide whether a clique of a given size exists in a given graph G . This is the CLIQUE decision problem which is known to be NP-complete [5, 6]. Thus, under the assumption that $P \neq NP$ no algorithm finds the minimizer of Eq. (17) in time polynomial in the binary encoding of the input vector $\mathbf{b}$.

To prove Theorem 1, we introduce several lemmas.

Lemma 2. *Let A be a non-zero, symmetric and circulant adjacency matrix of an undirected simple graph. This graph is d -regular with $d > 0$. Let α and β be real numbers with $\alpha \in [0, 1]$ and $\beta \in [0, \alpha/d]$. The non-negative coefficients b_n in Eq. (15) can be chosen such that*

$$M = \alpha \mathbb{I} + (1 - \alpha) \mathbb{J} + \beta A \quad (18)$$

holds, with $\mathbb{I}$ being the identity matrix and $\mathbb{J}$ the all-ones matrix.

Proof. It is evident from Eq. (14) that the matrix M is reversal invariant:

$$M_{0,k} = M_{0,N-k}, \quad (19)$$

which is a feature also shared by A

$$A_{0,k} = A_{k,0} = A_{0,N-k}.$$

The first equality follows by symmetry and the second by the circulant property, since $k - 0 \equiv 0 - (N - k) \pmod{N}$.

We prove the lemma by explicitly providing the coefficients b_n :

$$b_n = \alpha + N(1 - \alpha)\delta_{n,0} + \beta \sum_{l=0}^{N-1} A_{0,l} \cos\left(\frac{2\pi}{N} l n\right) \quad (20)$$

Inserting the form of b_n into Eq. (15), we obtain

$$\begin{aligned} M_{jk} &= \alpha \delta_{j,k} + 1 - \alpha + \beta \sum_l A_{0,l} \frac{1}{2} (\delta_{l,j-k} + \delta_{l,k-j}) \\ &= \alpha \delta_{j,k} + 1 - \alpha + \beta A_{j,k} \end{aligned} \quad (21)$$

where we used the circulant property of A and orthogonal relations for the cosines. Thus the chosen b_n lead to the desired matrix M .

It remains to show that the b_n are admissible, i.e. that they satisfy $b_n \geq 0$ and $\sum_n b_n = N$. The latter is straightforward to check: Summing over all n we obtain $\sum_n b_n = N$ because $A_{0,0} = 0$.

Since each row of a circulant adjacency matrix sums to the same value $d = \sum_l A_{0,l} > 0$, circulant graphs are d -regular. The value of d enters into the 0-component of b as

$$b_0 = \alpha + N(1 - \alpha) + \beta d. \quad (22)$$

Since $\alpha \in [0, 1]$ and $\beta \geq 0$ this is non-negative.

The remaining b_n with $n \neq 0$ satisfy

$$\begin{aligned} b_n &= \alpha + \beta \sum A_{0,l} \cos\left(\frac{2\pi}{N} l n\right) \\ &\geq \alpha - \beta \sum A_{0,l} = \alpha - \beta d \geq 0 \end{aligned} \quad (23)$$

since by assumption $\beta \in [0, \alpha/d]$. $\square$

In order to prove our hardness result, we draw from the work of [4]. We found this work by using ChatGPT Plus to assist us with literature discovery.

The authors of [4] utilize a theorem of Bose and Chowla [7] about the existence of a set of special positive integers.

Definition 3. Given an integer $n > 1$, let $N = 8^{\lceil \log_2 n \rceil} - 1$. We define a Bose-Chowla set as a collection of integers

$$\Lambda := \{a_1, \dots, a_n\}$$

that are distinct modulo $N + 1$ and have the following property: for all $i \leq j$ the two-term sums $a_i + a_j$ are distinct modulo N and for all $i \leq j \leq k$ all three-term sums $a_i + a_j + a_k$ are distinct modulo N .

Note that the elements themselves are required to be distinct modulo $N + 1$.

Given a graph $G = (V, E)$, the authors of [4] use the set Λ to construct a new circulant graph C that has the same maximum clique size.

Lemma 4. (Result of Codenotti, Gerace, Vigna [4])

Given a graph $G = (V, E)$ with $n = |V|$ vertices, define a new graph C with $N = 8^{\lceil \log_2 n \rceil} - 1 < 8n^3 - 1$ vertices and circulant adjacency matrix A . This matrix is fully specified by defining the first row:

$$A_{0,k} = \begin{cases} 1 & \text{if } k \equiv a_i - a_j \pmod{N} \text{ for some } (i, j) \in E \\ 0 & \text{otherwise.} \end{cases} \quad (24)$$

If the graph G has a clique of size K then the graph C has a clique of the same size and vice versa. The required Bose-Chowla set can be constructed in polynomial time in n . Thus the construction of C from G is accomplished in polynomial time.

In [4] this result was used to demonstrate NP-completeness of the CLIQUE decision problem which is defined as follows.

Input: A graph $G = (V, E)$ and a positive integer $K > 1$

Decide: Is there a clique of size K in G ?

In the later proof of our Theorem 1 we will reduce from this to our optimization problem.

We first turn to the construction of minimizers of the quartic term by using Bose-Chowla sets. The following lemma is based on the additive-energy minimization argument used in the proof of Proposition 4.29 of Tao and Vu [8]. We found this work by using ChatGPT Plus to assist us with literature discovery.

Lemma 5. Let G and C be the graphs from Lemma 4. We consider the problem of minimizing $C_4(z)$ with z constrained to having Hamming weight $H_0 \in [1, n]$:

$$\min_{z \in \{0,1\}^N} C_4(z) \quad (25)$$

$$\text{subject to } \sum_i z_i = H_0 \quad (26)$$

Pick any collection $\{i_1, \dots, i_{H_0}\}$ of H_0 vertices in G then the vector $z^C \in \{0, 1\}^N$ with components

$$z_a^C = \begin{cases} 1 & \text{if } a \in \Gamma := \{a_{i_1}, \dots, a_{i_{H_0}}\} \\ 0 & \text{otherwise} \end{cases} \quad (27)$$

is a minimizer of $C_4(z)$ with value

$$C_4(z^C) = \frac{16}{N} \sin^4\left(\frac{\gamma}{2}\right) (2H_0^2 - H_0). \quad (28)$$

Proof. Consider a binary vector $z \in \{0, 1\}^N$ with Hamming weight $\sum_i z_i = H_0$. We define the non-negative integer vector $\zeta \in \mathbb{Z}_{\geq 0}^N$ with components

$$\zeta_i := \sum_j z_j z_{j+i}. \quad (29)$$

Then the quartic term in Eq. (9) can be rewritten as

$$C_4(z) = \frac{16}{N} \sin^4\left(\frac{\gamma}{2}\right) \zeta \cdot \zeta. \quad (30)$$

In App. A we prove that

$$\zeta \cdot \zeta \geq 2H_0^2 - H_0. \quad (31)$$

We will now show that the ζ^C vector corresponding to z^C saturates this bound, thus z^C is a minimizer of $C_4(z)$. Firstly, note that for $i \neq 0$ we have $\zeta_i^C \in \{0, 1\}$. This can be deduced from the Bose-Chowla property of the a_i as follows. We consider

$$\zeta_i^C := \sum_j z_j^C z_{j+i}^C \quad (32)$$

for a fixed $i \neq 0$ and observe that in order to have $\zeta_i^C \geq 2$, there must exist two distinct values of j , say j_1 and j_2 , such that

$$z_{j_1}^C = z_{j_1+i}^C = z_{j_2}^C = z_{j_2+i}^C = 1. \quad (33)$$

From the definition Eq. (27) we deduce that either one of three possibilities occurs:

(a) There exist four distinct elements $a_1, a_2, a_3, a_4 \in \Gamma$ such that

$$j_1 \equiv a_1 \pmod{N} \quad (34)$$

$$j_2 \equiv a_2 \pmod{N} \quad (35)$$

$$j_1 + i \equiv a_3 \pmod{N} \quad (36)$$

$$j_2 + i \equiv a_4 \pmod{N} \quad (37)$$

thus

$$a_1 + a_4 \equiv a_3 + a_2 \pmod{N}. \quad (38)$$

By the element distinctness property of the two-term sums of a_i we deduce that either

$$a_1 \equiv a_3, a_2 \equiv a_4 \rightarrow i \equiv 0 \pmod{N} \quad (39)$$

or

$$a_1 \equiv a_2, a_3 \equiv a_4 \rightarrow j_1 = j_2 \pmod{N}, \quad (40)$$

both conclusions would contradict our assumptions about the indices.

(b) There exist three distinct elements $a_1, a_2, a_3 \in \Gamma$ such that

$$j_1 \equiv a_1 \pmod{N} \quad (41)$$

$$j_1 + i \equiv j_2 \equiv a_2 \pmod{N} \quad (42)$$

$$j_2 + i \equiv a_3 \pmod{N} \quad (43)$$

thus

$$a_1 + a_3 \equiv a_2 + a_2 \pmod{N}. \quad (44)$$

Again, by element distinctness of the two-term sums we deduce that

$$a_1 \equiv a_2 \equiv a_3 \rightarrow i \equiv 0, j_1 \equiv j_2 \pmod{N}, \quad (45)$$

contradicting our assumptions about the indices.

(c) There exist two distinct elements a_1, a_2 with

$$j_1 \equiv j_2 + i \equiv a_1 \pmod{N} \quad (46)$$

$$j_1 + i \equiv j_2 \equiv a_2 \pmod{N}. \quad (47)$$

However, this would imply $2i \equiv 0 \pmod{N}$, which reduces to $i \equiv 0 \pmod{N}$ since N is odd. This is contrary to our assumption about the indices. Thus we conclude that $\zeta_i^C \leq 1$ for $i \neq 0$.

Now with this information, we can calculate

$$\zeta \cdot \zeta = \zeta_0^2 + \sum_{i \neq 0} (\zeta_i^C)^2 = H_0^2 + \sum_{i \neq 0} \zeta_i^C = 2H_0^2 - H_0 \quad (48)$$

where we used that $\zeta_0^C = H_0$ and $\sum_i \zeta_i^C = H_0^2$, see App. A. Comparing with the lower bound Eq. (31), we see that $\mathbf{z}^C$ minimizes $C_4(\mathbf{z})$. $\square$

The next lemma shows that the form of the full cost function simplifies significantly for the representation in Lemma 2 when α is chosen suitably.

Lemma 6. Consider the cost function $C(\mathbf{z})$ after the matrix M is chosen as in Lemma 2. Then for

$$\alpha = \frac{4}{N} \sin^2 \left(\frac{\gamma}{2} \right) H_0 \quad (49)$$

the minimizers of $C(\mathbf{z})$ are identical to the minimizers of

$$C''(\mathbf{z}) = \frac{16}{N} \sin^4 \left(\frac{\gamma}{2} \right) \sum_i (\zeta_i + H_0 - H)^2 \quad (50)$$

$$-8 \sin^2 \left(\frac{\gamma}{2} \right) \beta \sum_i A_{0,i} (\zeta_i + H_0 - H). \quad (51)$$

Proof. We begin by choosing the b_n as in Lemma 2. Then

$$\sum_{j,k} M_{jk} z_j z_k = \alpha H + (1 - \alpha) H^2 + \beta \sum_{jk} A_{jk} z_j z_k \quad (52)$$

and the cost function takes the form

$$C' = \frac{16}{N} \sin^4 \left(\frac{\gamma}{2} \right) \sum_i (\zeta_i + H_0 - H)^2 - 8 \sin^2 \left(\frac{\gamma}{2} \right) \beta \sum_{jk} A_{jk} z_k z_j - 8 \sin^2 \left(\frac{\gamma}{2} \right) (\alpha H + (1 - \alpha) H^2) \quad (53)$$

$$+ 8 \sin^2 \left(\frac{\gamma}{2} \right) H^2 + 32 \sin^4 \left(\frac{\gamma}{2} \right) H H_0 - 8 \sin^2 \left(\frac{\gamma}{2} \right) (N - b_0) H - \frac{32}{N} \sin^4 \left(\frac{\gamma}{2} \right) H^2 H_0. \quad (54)$$

We have completed the square in the first sum in order to absorb the term that was cubic in H . The constant H_0 is a positive integer that will be chosen later. The cost function is denoted with prime because we have dropped a constant term involving H_0 .

Now, by choosing α as in the lemma's statement we can

make the quadratic order in H disappear and we are left with:

$$C' = \frac{16}{N} \sin^4 \left(\frac{\gamma}{2} \right) \sum_i (\zeta_i + H_0 - H)^2 \quad (55)$$

$$- 8 \sin^2 \left(\frac{\gamma}{2} \right) \beta \sum_i A_{0,i} (\zeta_i + H_0 - H) \quad (56)$$

$$- 8 \sin^2 \left(\frac{\gamma}{2} \right) \beta d H_0. \quad (57)$$

here we inserted the form of b_0 from Eq. (22) and rewrote the quadratic part in z in terms of ζ using

$$\sum_{jk} A_{jk} z_k z_j = \sum_i A_{0,i} \zeta_i. \quad (58)$$

Finally, the last term in Eq. (57) is a constant independent of z , thus we can drop it and arrive at the form stated in the lemma. $\square$

With this new form of the cost function in hand, the next lemma makes a statement about the first term.

Lemma 7. *Given an integer H_0 satisfying*

$$1 \leq H_0 \leq \min(n, (N+1)^{1/3} - 1), \quad (59)$$

the function

$$f(z) = \sum_i (\zeta_i + H_0 - H)^2 \quad (60)$$

is bounded from below by

$$f(z) \geq 2H_0^2 - H_0. \quad (61)$$

The inequality is strict unless $H \in \{H_0, N - H_0\}$, in which case equality can be achieved.

Proof. $f(z)$ has the property that it remains invariant if we replace z by the complementary vector $\bar{z}$ defined as

$$\bar{z} := \mathbf{1} - z \quad (62)$$

where $\mathbf{1} = (1, 1, \dots, 1)$ is the all-ones vector.

To see this, note that

$$\bar{\zeta}_i := \sum_j \bar{z}_j \bar{z}_{j+i} = N - 2H + \zeta_i \quad (63)$$

and

$$\bar{H} := \sum_j \bar{z}_j = N - H. \quad (64)$$

Thus

$$\bar{\zeta}_i - \bar{H} = \zeta_i - H \quad (65)$$

and we conclude

$$f(z) = f(\bar{z}). \quad (66)$$

Thus, in the following we can confine ourselves to z with $H \leq N/2$ and keep in mind that if z_0 is a minimizer of $f(z)$ so is $\mathbf{1} - z_0$.

Now we prove inequality (61). Consider first the case $H < H_0$. Here we have

$$f(z) \geq \sum_i (\zeta_i + 1)^2 \geq N \quad (67)$$

$$\geq (H_0 + 1)^3 - 1 > 2H_0^2 - H_0. \quad (68)$$

The case $H > H_0$ is more difficult. We first separate off $\zeta_0 = H$ from the sum and use the Cauchy-Schwarz inequality on the remaining terms to obtain

$$f(z) = H_0^2 + \sum_{i \neq 0} (\zeta_i + H_0 - H)^2 \quad (69)$$

$$\geq H_0^2 + \frac{[H(N - H) - (N - 1)H_0]^2}{N - 1}. \quad (70)$$

Then we note that the quadratic form in H is non-negative

$$H(N - H) - (N - 1)H_0 \geq 0 \quad (71)$$

for $H \in [H_0 + 1, N/2]$. It is also a strictly increasing function in that same range. Thus we can bound this term by its value at $H = H_0 + 1$ and obtain

$$f(z) \geq H_0^2 + \frac{[N - 1 - H_0 - H_0^2]^2}{N - 1} > 2H_0^2 - H_0 \quad (72)$$

where in the last inequality we used $N \geq (H_0 + 1)^3 - 1$.

Finally, for the case $H = H_0$ we can use the construction of z^C from Lemma 5 since $H = H_0 \leq n$. We obtain

$$f(z^C) = \sum_i (\zeta_i^C)^2 = 2H_0^2 - H_0 \quad (73)$$

thereby saturating the bound in Eq. (31). The complementary vector $\bar{z}^C$ yields the same value for f . $\square$

We are finally in a position to prove Theorem 1.

Proof of Theorem 1. The proof works by showing that we can reduce the decision problem CLIQUE in polynomial time to an instance of the optimization problem Eq. (7).

Let $G = (V, E)$ be a given simple graph and $K > 1$. We assume that the edge set E is not empty since otherwise the CLIQUE answer is NO, trivially. We may also assume that $K \leq |V|$ since otherwise the answer to CLIQUE is again NO.

For later convenience, we assume without loss of generality that G has an isolated vertex. This does not change whether G has a clique of size $K > 1$. Let $n = |V|$ denote the number of vertices after this preprocessing.

We use the construction of Lemma 4 to obtain a new graph C with $N < 8n^3 - 1$ vertices. This is accomplished in polynomial time according to Lemma 4. Let A be the adjacency matrix of C defined there. In the optimization problem in Eq. (7) we choose the matrix M according to Eq. (18) of Lemma 2. Choosing α as in Lemma 6, the optimization problem Eq. (7) with matrix M is equivalent to minimizing

$$C''(z) = \frac{16}{N} \sin^4\left(\frac{\gamma}{2}\right) \sum_i (\zeta_i + H_0 - H)^2 \quad (74)$$

$$- 8 \sin^2\left(\frac{\gamma}{2}\right) \beta \sum_i A_{0,i} (\zeta_i + H_0 - H). \quad (75)$$

As noted in Lemma 7, $C''(z)$ is invariant under $z \mapsto \mathbf{1} - z$, since $\zeta_i + H_0 - H$ is invariant under this transformation. Hence, without loss of generality we restrict our attention to vectors with $H \leq N/2$.

The value of β has to be in the range $[0, \alpha/d]$. Since the first term is an integer multiple of $16/N \sin^4 \gamma/2$, its minimum value and next higher value have a gap $\Delta > 0$ between them. This gap has to be an integer multiple of $16/N \sin^4 \gamma/2$, in particular

$$\Delta \geq \frac{16}{N} \sin^4 \frac{\gamma}{2}. \quad (76)$$

The sum in Eq. (75) can be bounded by the triangle inequality:

$$\begin{aligned} \left| \sum_i A_{0,i}(\zeta_i + H_0 - H) \right| &\leq \sum_i (\zeta_i + |H_0 - H|) \quad (77) \\ &\leq \sum_i (N + N) = 2N^2 \quad (78) \end{aligned}$$

Moreover, the construction in Lemma 7 provides a minimizer of the first term for which the sum appearing in the second term is non-negative. Thus, if β is sufficiently small, the maximal possible decrease of the second term cannot compensate for the gap Δ between the minimum and the next possible value of the first term. Thus, when

$$\beta < \frac{1}{N^3} \sin^2 \frac{\gamma}{2} \quad (79)$$

the minimizer of $C''(z)$ is a subset of minimizers of the first term. Now choose a β that satisfies condition (79), say

$$\beta = \frac{1}{2N^3} \sin^2 \frac{\gamma}{2}. \quad (80)$$

Then the minimizers of the cost function $C''(z)$ are a subset of the minimizers of the first term (74) by itself.

The choice of α in Eq. (49) and β in Eq. (80) are admissible in Lemma 2. For β we see this since $\beta d/\alpha = d/(8H_0 N^2) < 1$ where we used $d < N$. For α , we note that

$$\alpha = \frac{4}{N} \sin^2 \frac{\gamma}{2} H_0 \leq \frac{4(n-1)}{n^3-1} = \frac{4}{n^2+n+1} < 1 \quad (81)$$

for $n \geq 2$.

We show now that we can solve the CLIQUE decision problem on graph G by solving the problem of optimizing C'' . First we set $H_0 = K$, which is admissible because

$$K \leq n-1 \leq (N+1)^{1/3} - 1. \quad (82)$$

The first inequality holds because G contains an isolated vertex. The second inequality follows since $N \geq n^3 - 1$.

According to Lemma 7 the first term, Eq. (74), of $C''(z)$ has a lower bound $16/N \sin^4(\frac{\gamma}{2})(2H_0^2 - H_0)$. By Lemma 7, with the restriction $H \leq N/2$, this lower bound cannot be attained for $H \neq H_0 = K$.

At $H = H_0$ the first term, Eq. (74), in C'' is identical to $C_4(z)$. We have shown in Lemma 5 that we can construct z^C vectors with $C_4(z^C) = 16/N \sin^4(\frac{\gamma}{2})(2H_0^2 - H_0)$, saturating the lower bound. Thus minimizers exist that saturate the lower bound of $C''(z)$ and they all have Hamming weight $H = H_0 = K$.

If there is a clique of size $H_0 = K$ in G let $I := (i_1, \dots, i_K)$ denote its vertices. According to Lemma 5 we can construct a binary vector z^C from the vertices I that minimizes the first sum in C'' . Now we show that this vector also minimizes the second term in C'' . This latter term reads

$$-8 \sin^2 \left(\frac{\gamma}{2} \right) \beta \sum_i A_{0,i} \zeta_i \quad (83)$$

$$= -8 \sin^2 \left(\frac{\gamma}{2} \right) \beta \sum_{j,k} A_{j,k} (z^C)_j (z^C)_k \quad (84)$$

where we have set $H = H_0 = K$ since this is enforced by the first term. Recall that z^C is supported on indices in Γ , see Eq. (27).

Since I is a clique in G the set $\Gamma = \{a_{i_1}, \dots, a_{i_K}\}$ is a clique in C . This is easy to see: let $i, i' \in I$ then $(i, i') \in E$. C has an adjacency matrix which has a 1 in position $(0, a_i - a_{i'})$. By the circulant property it has a 1 in position $(a_{i'}, a_i)$. Thus the vertices $a_{i'}$ and a_i in C are joined. In other words, if I forms a clique of size K in G then Γ forms a clique of size K in C .

Therefore, in Eq. (84) the sum can be evaluated and yields

$$-8 \sin^2 \left(\frac{\gamma}{2} \right) \beta K(K-1). \quad (85)$$

This is the lowest possible value of $C''(z)$ since the Hamming weight of z^C is K . Thus, when the answer to CLIQUE is YES the optimization yields the value

$$\begin{aligned} C''_{\text{YES}} &:= C''(z^C) \quad (86) \\ &= \frac{16}{N} \sin^4 \left(\frac{\gamma}{2} \right) (2K^2 - K) - 8 \sin^2 \left(\frac{\gamma}{2} \right) \beta K(K-1). \end{aligned}$$

Now we show that if the answer is NO, the optimization yields a higher value of C'' . According to Lemma 4, if there is no clique of size K in G , there cannot be a clique of size K in C . Thus when the decision problem has answer NO, the sum in Eq. (84) assumes a value that is at most $K(K-1) - 2$. This is because the minimizing z has Hamming weight K and A is symmetric with zero diagonal.

Thus a minimizer z_{opt} of $C''(z)$ will yield the same value as before for the first term, Eq. (74), but will yield a value for the second term, Eq. (84), that is larger than in the YES case:

$$C''_{\text{NO}} := C''(z_{\text{opt}}) > C''_{\text{YES}}. \quad (87)$$

Thus, the YES and NO answers are separated by an inverse-polynomial-size gap

$$C''_{\text{NO}} - C''_{\text{YES}} \geq 16 \sin^2 \left(\frac{\gamma}{2} \right) \beta = \frac{8}{N^3} \sin^4 \left(\frac{\gamma}{2} \right). \quad (88)$$

In other words, by solving the optimization problem we are able to decide from the value of C'' if there is a clique of size K or not. If there is, the optimization yields a value equal to Eq. (86). If not, we obtain a value that is larger than this by at least an amount equal to Eq. (88).

As a final part of the proof, it remains to be shown that the reduction is accomplished in polynomial time. The encoding

of the matrix M and the generation of the Bose-Chowla set can be done in polynomial time [4]. However, the intensity vector $\mathbf{b}$ calculated from Eq. (20) will in general be irrational. In order to encode it in polynomial time we form a rational approximation $\hat{\mathbf{b}} \in \mathbb{Q}^N$ to $\mathbf{b}$. Let us approximate component-wise for $i \neq 0$ by

$$|\hat{b}_i - b_i| < \frac{\epsilon}{N-1}. \quad (89)$$

Then define

$$\hat{b}_0 = N - \sum_{i \neq 0} \hat{b}_i. \quad (90)$$

These steps can be accomplished in polynomial time in $1/\epsilon$ from Eq. (20) since $\cos 2\pi n/N$ can be approximated to d bit accuracy in polynomial time in d , see [9]. For sufficiently small ϵ the vector $\hat{\mathbf{b}}$ will be non-negative.

Note that we have the bound

$$|\hat{b}_0 - b_0| = \left| \sum_{i \neq 0} \hat{b}_i - b_i \right| \leq \sum_{i \neq 0} |\hat{b}_i - b_i| < \epsilon. \quad (91)$$

We will now bound the change in the cost function when $\mathbf{b}$ is replaced by $\hat{\mathbf{b}}$. The only places in the cost function where the $\mathbf{b}$ vectors appear are in Eq. (16) and the matrix M in Eqs. (14)-(15). Let us denote the cost function with $\hat{\mathbf{b}}$ in place of $\mathbf{b}$ by $\hat{C}(z)$. Similarly, let $\hat{M}$ denote the matrix corresponding to $\hat{\mathbf{b}}$. Then we have

$$\begin{aligned} |\hat{M}_{jk} - M_{jk}| &= \frac{1}{N} \left| \sum_{n=0}^{N-1} (\hat{b}_n - b_n) \cos \left(\frac{2\pi(j-k)n}{N} \right) \right| \\ &\leq \frac{1}{N} \sum_{n=0}^{N-1} |\hat{b}_n - b_n| < \frac{2\epsilon}{N} < \epsilon. \end{aligned} \quad (92)$$

The cost function itself will change by an amount that we can upper bound:

$$\begin{aligned} |\hat{C}(z) - C(z)| &\leq 8 \sin^2 \frac{\gamma}{2} \left| \sum_{jk} [\hat{M}_{jk} - M_{jk}] z_j z_k \right| \\ &\quad + 8 \sin^2 \frac{\gamma}{2} |(\hat{b}_0 - b_0) H| \\ &< 8 \sin^2 \frac{\gamma}{2} (\epsilon H^2 + \epsilon H) \end{aligned} \quad (93)$$

$$\leq 8 \sin^2 \frac{\gamma}{2} (N^2 + N) \epsilon. \quad (94)$$

Since C and C'' only differ by a z -independent constant, the same bound applies to C'' . We denote by $\hat{C}''$ the cost function obtained from C'' after rational approximation of $\mathbf{b}$. If we now carry out the rational approximation, the gap between the NO and YES answer of Eq. (88), is at most changed to

$$\hat{C}''_{\text{NO}} - \hat{C}''_{\text{YES}} \geq \frac{8}{N^3} \sin^4 \left(\frac{\gamma}{2} \right) - 16 \sin^2 \frac{\gamma}{2} (N^2 + N) \epsilon. \quad (95)$$

Here we used Eq. (94) and the triangle inequality. Thus by choosing

$$\epsilon = \frac{1}{4N^3(N^2 + N)} \sin^2 \left(\frac{\gamma}{2} \right) \quad (96)$$

the gap between the NO and YES answers remains positive and an inverse polynomial in N . As noted above, the approximation in Eq. (89) can be accomplished in polynomial time in $1/\epsilon$, i.e. in polynomial time in N . Since $N < 8n^3 - 1$, it can therefore be accomplished in polynomial time in n .

Thus we have reduced CLIQUE in polynomial time to the optimization problem Eq. (17). This finishes the proof of Theorem 1. $\square$

IV. CONCLUSION

We have studied the problem of designing a binary phase mask and proved that the associated optimization problem is NP-hard. The result holds despite having a fixed Fourier operator and despite the restriction to physically admissible target intensity vectors. This shows that the discrete nature of the phase settings is sufficient to produce computationally hard instances.

Our problem can also be viewed as a particular version of the phase retrieval problem. In the design setting that we studied in this paper, the uniqueness of the solution does not play a role. Instead, any phase mask setting that minimizes the distance to the target pattern constitutes an optimal solution.

Our result concerns worst-case exact optimization and does not preclude the possibility that efficient algorithms may exist for particular subclasses of intensity vectors. Therefore, an important direction for future work is to identify structured classes of intensity vectors for which efficient optimization algorithms exist. It would also be interesting to investigate how the computational hardness changes for phase masks in higher-dimensional optical settings.

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Appendix A: Bounds for the quartic term

In the quartic term $C_4(\mathbf{z})$ of the main text, the sum can be bounded in terms of the Hamming weight $H = \sum_i z_i$ as follows:

$$2H^2 - H \leq \sum_{j,k,l} z_k z_{j+k} z_l z_{j+l} \leq H^3 \quad (\text{A1})$$

Proof. We introduce the auto-correlation vector $\zeta := (\zeta_0, \dots, \zeta_{N-1})$ with components

$$\zeta_i := \sum_l z_l z_{l+i} \quad (\text{A2})$$

and rewrite the quartic sum in question as the inner product

$$\sum_{j,k,l} z_k z_{j+k} z_l z_{j+l} = \zeta \cdot \zeta. \quad (\text{A3})$$

We have $\zeta_i = \sum_l z_l z_{l+i} \leq \sum_l z_l 1 = H$. This inequality is saturated by

$$\zeta_0 = \sum_l z_l = H. \quad (\text{A4})$$

Also, note that

$$\sum_i \zeta_i = \sum_{i,l} z_l z_{l+i} = \left(\sum_j z_j \right)^2 = H^2. \quad (\text{A5})$$

Using these facts, we can derive the upper bound:

$$\zeta \cdot \zeta = \sum_i \zeta_i \zeta_i \leq \max_i \zeta_i \cdot \sum_i \zeta_i = H^3 \quad (\text{A6})$$

To prove the lower bound, we note that $\zeta_i^2 \geq \zeta_i$ holds, thus

$$\sum_l \zeta_l^2 = \zeta_0^2 + \sum_{l \neq 0} \zeta_l^2 \geq H^2 + \sum_{l \neq 0} \zeta_l = 2H^2 - H, \quad (\text{A7})$$

where we used Eqs. (A4) and (A5). In the main text we show that the lower bound can be saturated by selecting a binary vector based on the special set of integers Λ . $\square$

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